

Operation Emberlight — Student Cover Sheet

VCE Algorithmics (HESS) SAT 2026

Student name:

Student number:

School:

Teacher:

Year:

2026

Your Facility Seed

SEED: _____

This number uniquely identifies your facility layout.
It must appear in the first cell of your Marimo notebook.
Keep it safe — it cannot be reissued.

Submission Checklist

Component	Submitted	Teacher initials
☐ Memo 01 response	_____	_____
☐ Memo 01 notebook	_____	_____
☐ Memo 02 response	_____	_____
☐ Memo 02 notebook	_____	_____
☐ Memo 03 response	_____	_____
☐ Memo 03 notebook	_____	_____
☐ Memo 04 response	_____	_____
☐ Memo 04 notebook	_____	_____
☐ Memo 05 response	_____	_____
☐ Memo 05 notebook	_____	_____

Declaration: I declare that all resource materials and assistance used in this task have been acknowledged, and that all unacknowledged work is my own. I understand that my Marimo notebook must reflect my own progressive thinking and that its commit history may be reviewed for authentication purposes.

Signature: _____

Date: _____



Duration: 12 weeks

Style: Adaptable Design Problem

Deliverable: Design → Analyse → Improve → Compare

Situation Report

Three days ago, seismic activity destabilised the **Emberlight Subterranean Research Complex**. Automated containment systems are failing. Critical supply units remain scattered throughout the structure. Human entry is no longer possible. You have been assigned to the **Autonomous Systems Squad (ASS)**. Your task is to develop the **control strategy** for a rescue drone operating inside the complex.

Mission Directive

The Rescue Authority defines success as:

Maximise recovered priority while **maintaining structural integrity** and **ensuring successful extraction**.

They do *not* prescribe a method. They *expect* a strategy.

Your Role

You are not writing code for a simple robot.
You are designing a **decision architecture**.

You must:

1. decide how the facility and mission should be represented computationally;
2. identify the constraints governing the drone's operation;
3. design an initial strategy and communicate it clearly;
4. analyse efficiency and feasibility;
5. refine and improve your design as new constraints emerge;
6. justify decisions and compare alternatives with depth.



Final Statement: This is a design problem, not a “follow-the-steps” exercise. You will receive updates over the 12 weeks. Deep modelling, formal reasoning, meaningful improvement, and clear comparison are what make a strong submission. **Work must be completed and submitted in class each week to ensure you achieve a grade for each criterion.**

Facility Snapshot



A small example layout (for theme only).
Your actual layouts will vary.

Unlock Badges

Side quests you *may* activate later:

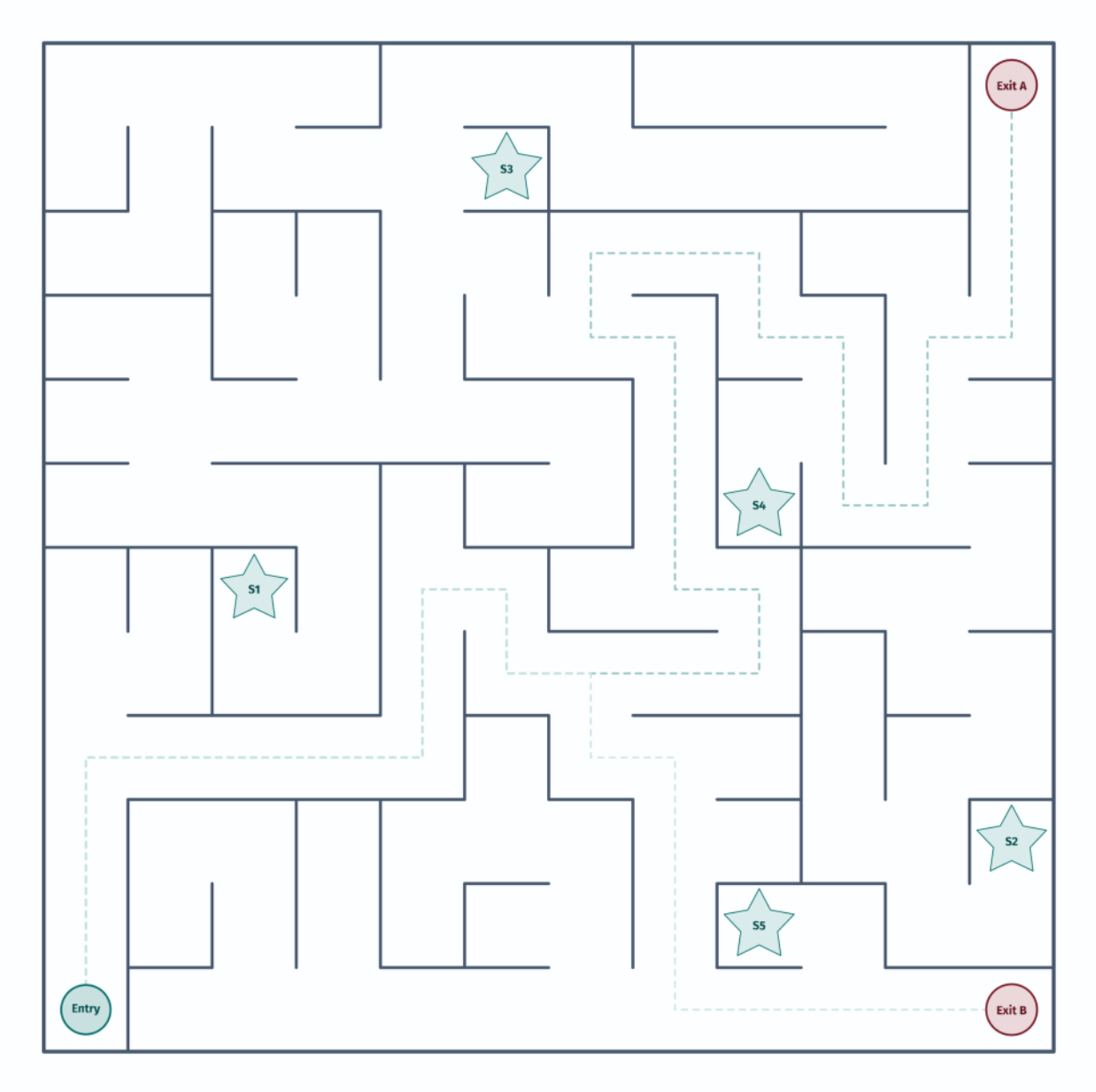
- E** Multi-exit extraction
- W** Variable corridor strain
- L** Load-sensitive energy cost
- ?** Uncertainty / fog of war
- R** Risk gradients / decay

Badges add complexity *over time* — not all at once.

Operational Phases

- I** Stable
Uniform traversal
- II** Instability
Variable cost
- III** Load
Cost depends on load
- IV** Refine
Improved model + compare

Facility Layout: Sector Grid Zone A



Classification: RESTRICTED — Autonomous Systems Squad (ASS) only

Issued by: Emberlight Rescue Authority, Engineering Division

Subject: Initial deployment authorisation for Rescue Drone Unit 1

Situation Summary

At 03:47 on Day 1, seismic activity registering 6.2 on the Richter scale struck the Emberlight Subterranean Research Complex. Automated containment systems have partially failed. Emergency lighting is operational. Five critical supply units — designated **S1 through S5** — remain scattered throughout the structure and must be recovered before secondary collapse renders the lower sectors inaccessible.

Human entry has been ruled out. You have been assigned command of **CRUDY-1** (Corridor Reconnaissance and Utility Drone — Year 1), a semi-autonomous aerial platform capable of navigating the facility's corridor network and transporting recovered supply units to an extraction point.

Your mission is not to write code for a simple robot. You are designing a **decision architecture** — a formal model of the problem and a strategy for solving it.

Facility Technical Briefing

The Emberlight Complex is organised as a **sector grid**. Each sector is a discrete navigable unit. Sectors are connected by reinforced corridors. The following has been confirmed by post-seismic structural inspection:

- The facility is **fully connected** — every sector is reachable from every other sector via some sequence of corridors.
- Each corridor connects exactly **two adjacent sectors**. Diagonal movement is not possible.
- All corridors are **bidirectional** — CRUDY-1 may traverse any open corridor in either direction.
- All navigable corridors are currently **structurally equivalent**. Traversal strain is uniform across all transitions.
- Movement between adjacent sectors incurs **one unit of energy expenditure**, regardless of direction.
- **No dynamic degradation** is expected during initial deployment. Conditions are stable.

Your printed facility schematic identifies the following:

- **Entry point** — CRUDY-1 deployment sector (bottom-left).
- **Exit A** — Primary extraction point (top-right).
- **Exit B** — Secondary extraction point (bottom-right).
- **Supply units S1-S5** — scattered throughout the grid.

Operational Constraints

The Rescue Authority has confirmed the following hard constraints governing CRUDY-1's operation. Your solution **must** satisfy all of them.

Load capacity	CRUDY-1 can carry a maximum of 5 supply units simultaneously. All supply units have uniform mass (weight = 1).
Extraction	CRUDY-1 must reach either Exit A or Exit B to complete the mission. It may not terminate mid-facility.
Energy budget	No hard energy limit is imposed at this stage. However, your strategy should aim to minimise total traversal cost where possible.
Revisiting sectors	CRUDY-1 <i>may</i> revisit sectors.
Supply collection	A supply unit is collected when CRUDY-1 occupies its sector. Collection is instantaneous. Each unit can only be collected once.
Objective	Maximise recovered supply priority while ensuring safe extraction . Recovery of all five units is preferred but not always feasible given energy constraints.

Mission Directive

Design a formal model of the Emberlight facility and develop an initial control strategy for CRUDY-1. Your strategy must be expressed precisely — first as a data model using appropriate Abstract Data Types, then as a designed algorithm, then in pseudocode, and finally justified in writing.

The Rescue Authority does *not* prescribe a method. They expect a **reasoned, well-communicated strategy** that can be formally analysed.

Required Actions

Complete each of the following in your Marimo workbook. Responses must be written in the designated text cells. Pseudocode must use standard Algorithmics conventions.

Action 1 — Problem Specification

Criterion 1 · Observation 1

Precisely specify the algorithmic problem formulated from this scenario. Your specification must address:

- **Inputs:** What data does the algorithm receive? Define these formally using your ADTs.
- **Output:** What does a correct solution look like? What does CRUDY-1 return?
- **Constraints:** Which of the operational constraints above are modelled in your problem specification?
- **Salient features:** Which real-world features of this problem must be captured in the model? Justify your selection.
- **Data model:** Identify the ADTs you will use. For each, describe what it models, why it is appropriate, and provide signatures for its key operations.

No fixed word count — this is a design artefact. Precision and justification are assessed, not length.

Action 2 — Algorithm Design

Criterion 2 · Observation 2 · 600–800 words

Design an algorithmic solution for CRUDY-1's control strategy. Your response must:

- Consider **at least two** algorithm design approaches (e.g. uninformed search, informed search, greedy strategies). For each, discuss its suitability and limitations for this problem.
- Clearly explain the approach you **selected** and the rationale underpinning that selection.
- Describe how your algorithm handles the core challenge: *balancing supply collection against extraction safety*.

Word range: approximately 600–800 words.

Action 3 — Pseudocode

Criterion 3 · Observation 3

Communicate your algorithm in pseudocode in your Marimo notebook. Your pseudocode must:

- Accurately reflect your algorithm design from Action 2.
- Use a **modular approach** — define sub-functions or procedures where appropriate.
- Reference your **ADT operations** from Action 1 (e.g. `neighbours(v)`, `enqueue(q, v)`).
- Correctly use iteration, conditionals, and recursion where appropriate.

Pseudocode goes in a text cell in your notebook, not as code comments.

Action 4 — Justification

Criterion 4 · Observation 4 · 300–500 words

Justify your chosen solution. Your justification must clearly demonstrate:

- **Suitability** — why this algorithm is appropriate for the Emberlight problem.
- **Coherence** — how your algorithm and data model work together as an integrated solution.
- **Fitness for purpose** — how your solution satisfies the mission directive and operational constraints above.

Word range: approximately 300–500 words.

Submission

Submit your completed Marimo notebook at **Observation 4**. Ensure your facility seed appears in the first cell of the notebook and that all four actions are addressed.

Work must be developed progressively in class. Teachers will observe and initial your authentication record at each observation point.

This memo remains active for the duration of Unit 3. Subsequent memos will introduce new constraints — your model will need to evolve accordingly.

Proceed.